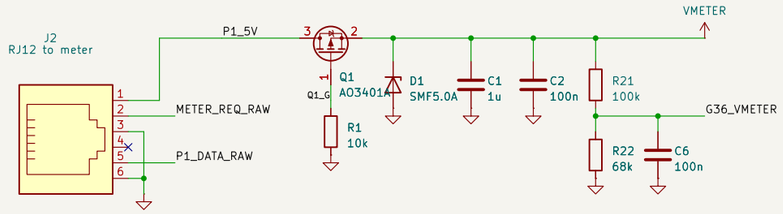
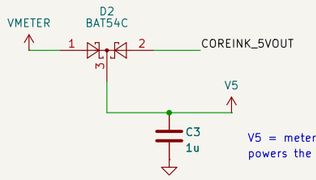
1 Meter connector, reverse-polarity protection, supplies



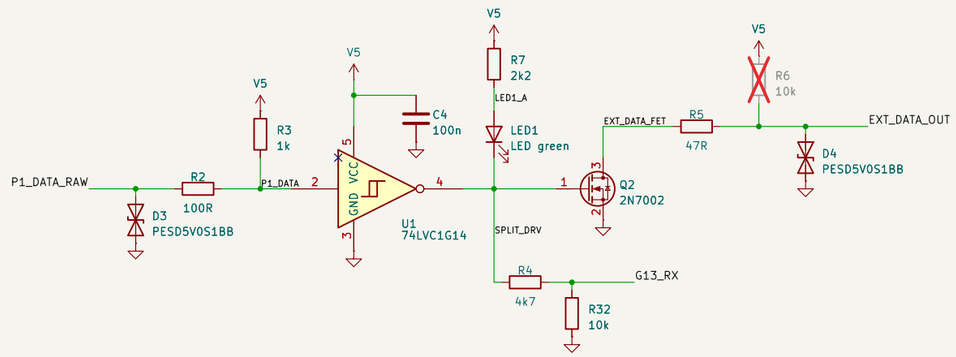
To the smart meter (P1 port).

Q1: reverse-polarity protection (conducts only with the meter's 5 V on pin 1). D1: surge clamp. G36 reads the meter voltage (x 0.405).



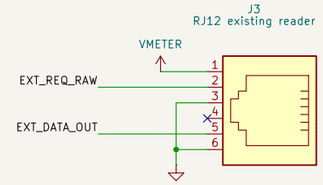
V5 = meter 5 V or CoreInk 5V_OUT, whichever is higher; powers the input, the splitter and the data request.

2 P1 data input, Schmitt trigger, active splitter



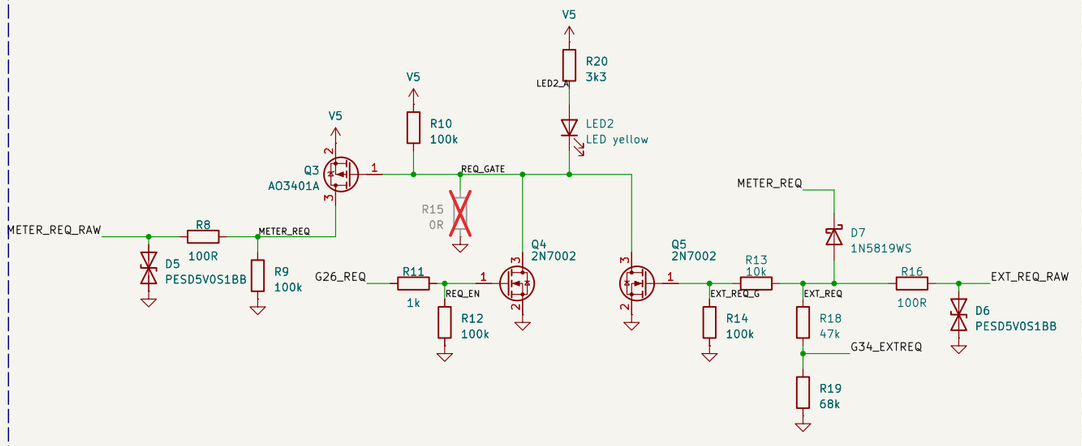
Schmitt trigger at V5 with the 1 k pull-up of v2.7 (reads weak meter outputs).
Q2 drives the existing reader from V5, so it keeps working when the CoreInk is off.
R6 (not fitted): pull-up for readers without their own.
R4/R32: 5 V to 3.2 V for the ESP32 (G13 = P1 RX, not inverted).
LED1 (green): blinks with the meter data.

3 Existing P1 reader



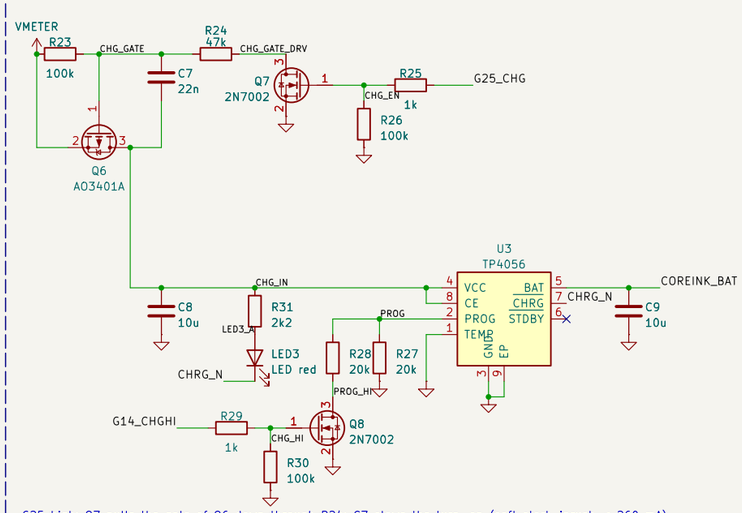
For a reader that was already on the meter (HomeWizard P1 meter, ...): it gets the meter's 5 V and a buffered copy of the data, and its data request is passed on to the meter.

4 Data request: CoreInk (G26) OR existing reader



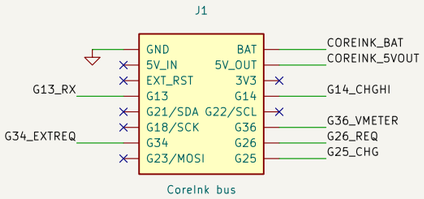
The meter sends when its request pin (J2 pin 2) is >= 4 V. Q3 pulls it to V5 when REQ_GATE is low: Q4 when the CoreInk requests (G26 high), Q5 when the existing reader requests. D7 passes the reader's own request on when V5 is absent (DSMR 2/3 meter with the CoreInk off). G34 tells the firmware that the reader requests. R15 (not fitted): always request. LED2 (yellow): data requested.

5 P1 charging of the CoreInk battery



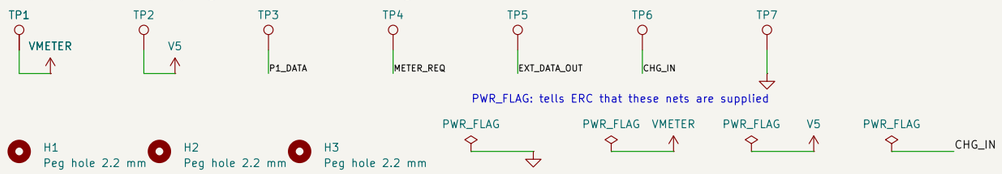
G25 high: Q7 pulls the gate of Q6 down through R24; C7 slows the turn-on (soft start, inrush < 260 mA).
TP4056 charges the CoreInk battery: 60 mA (R27), 120 mA with G14 high (R27 || R28).
LED3 (red): charging.

6 CoreInk bus (back connector)



Firmware interface (CoreInk ESP32):
G13 P1 data in (UART RX, not inverted)
G26 request data (high = request)
G34 existing reader requests (in)
G36 meter 5 V x 0.405 (ADC)
G25 P1 charging on (high)
G14 charge current 120 mA (high) / 60 mA
BAT CoreInk battery (charged by U3)
5V_OUT CoreInk 5 V, feeds V5 via D2

7 Test points, peg holes, ERC flags



PWR_FLAG: tells ERC that these nets are supplied

H1 Peg hole 2.2 mm
H2 Peg hole 2.2 mm
H3 Peg hole 2.2 mm

Revision history

2.7 2024-06 Windesheim: optocoupler splitter, DIP switches for the data request.
3.0 2026-11 Active splitter, firmware-managed data request, P1 charging, all parts top side.

Design rationale, calculations and simulations: docs/design-v3.md.
DNP = not fitted (R6, R15).

Continues the P1-BASE by Research group Energy Transition, Windesheim
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M5Stack CoreInk extension: P1 smart meter reader with active splitter
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Title: NeedForHeat P1-BASE

Size: A3	Date: 2026-11-14	**Rev: 3.0**
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